LIST OF PUBLICATIONS

ALEX ELZENAAR

Preprints and submitted

[1] A. Elzenaar, Quasi-Fuchsian groups and complex realisations of q-deformed real numbers, 2026. arXiv 2609.29831 [math.CV].

Abstract: We relate the theory of q-rational and q-real numbers introduced by Morier-Genoud and Ovsienko to the classical theory of Kleinian groups and their Teichmüller spaces. This provides a geometric point of view on several recent results about realisations of q-rationals for particular values of $q \in \mathbb{C}$. As an application we resolve a conjecture of Bapat, Becker, and Licata on the topology of the q-deformed Farey tessellation.

[2] A. Elzenaar, On rank two Kleinian groups with three parabolics, 2026. arXiv 2607.07424 [math.GR].

Abstract: The free group of rank 2 is the fundamental group of the genus 2 handlebody $\mathcal{H}$. We study discrete representations of this group into $\mathsf{PSL}(2,\mathbb{C})$ so that three disjoint simple closed curves on the conformal boundary $\partial_\infty \mathcal{H}$ are sent to parabolic elements. We show that the only infinite covolume groups of this form are maximal cusp groups on the boundary of genus 2 Schottky space. We also exhibit hitherto unexpected finite covolume groups which do not arise from Heegaard splitting presentations of tunnel number 1 links.

[3] A. Elzenaar, Expansion joints in hyperbolic manifolds, 2025, Updated Jun 2026. arXiv 2512.00879 [math.GT].

Abstract: Deformations of hyperbolic manifolds through metrics with cone singularities along closed loops were first studied by Thurston as continuous realisations of Dehn fillings. Instead of gluing singular solid tori into rank 2 cusps, we glue singular 2-handles into rank 1 cusps. To do this we find substructures within which the hyperbolic metric can be 'fractured' in a controlled way by direct manipulation of a fundamental polyhedron, changing the cone angle around an ideal arc to interpolate between cusped hyperbolic manifolds and hyperbolic manifolds with conformal surfaces on the visual boundary. As an application, we use cone deformations of a family of arithmetic manifolds derived from the Borromean rings to show that the upper unknotting tunnels of highly twisted 2-bridge links can be drilled out by cone deformations through pinched negatively curved metrics. Finally we show that our structures arise naturally in fully augmented links, providing a large family of examples.

[4] A. Elzenaar, Peripheral subgroups of Kleinian groups, 2025, Updated Dec 2025. arXiv 2508.00297 [math.GT].

Abstract: The conformal boundary of a hyperbolic 3-manifold M is a union of Riemann surfaces. If any of these Riemann surfaces has a nontrivial Teichmüller space, then the hyperbolic metric of M can be deformed quasi-isometrically. These deformations correspond to small pertubations in the matrices of the holonomy group $\pi_1(M) \subset \mathsf{PSL}(2,\mathbb{C})$, which together give an island of discrete representations around the identity map in $X = \mathrm{Hom}(\pi_1(M), \mathsf{PSL}(2,\mathbb{C}))$. Determining the extent of this island is a hard problem. If M is geometrically finite and its convex core boundary is pleated only along simple closed curves, then we cut up its conformal boundary in a way governed by the pleating combinatorics to produce a fundamental domain for $\pi_1(M)$ that is combinatorially stable under small deformations, even those which change the pleating structure. We give a computable region in X, cut out by polynomial inequalities over $\mathbb{R}$, within which this fundamental domain is valid: all the groups in the region have peripheral structures that look 'coarsely similar', in that they come from real-algebraically deforming a fixed conformal polygon and its side-pairings. The union of all these regions for different pleating laminations gives a countable cover, with sets of controlled topology, of the entire quasi-isometric deformation space of $\pi_1(M)$—which is known to be topologically wild.

[5] A. Elzenaar, From disc patterns in the plane to character varieties of knot groups, 2025, Updated Aug 2025. arXiv 2503.13829 [math.GT].

Abstract: Motivated by an experimental study of groups generated by reflections in planar patterns of tangent circles, we describe some methods for constructing and studying representation

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spaces of holonomy groups of infinite volume hyperbolic 3-manifolds that arise from unknotting tunnels of links. We include full descriptions of our computational methods, which were guided by simplicity and generality rather than by being particularly efficient in special cases. This makes them easy for non-experts to understand and implement to produce visualisations that can suggest conjectures and support algebraic calculations in the character variety. Throughout, we have tried to make the exposition clear and understandable for graduate students in geometric topology and related fields.

[6] A. Elzenaar, Changing topological type of compression bodies through cone manifolds, 2024, Updated Mar 2026. arXiv 2411.17940 [math.GT].

Abstract: Homeomorphism types of compression bodies form the vertices of a graph where two vertices are joined by an edge if one compression body is obtained by gluing a 2-handle onto the other. Motivated by earlier work of Lackenby and Purcell on geodesicity of unknotting tunnels for hyperbolic links, we show that it is possible to realise all of the edges in the graph of compression bodies by paths of cone manifold holonomy groups such that the handle that is glued in is obtained as a limit of singular arcs of cone angle increasing from 0 to 2π. We apply standard techniques from the theory of CAT(0) spaces, and do not rely on the harmonic deformation theory of Hodgson and Kerckhoff. Along the way we prove a generalisation of a classic theorem of Koebe and Maskit on existence of function groups which implies existence results for reflex angled hyperbolic cone structures on a wide range of compression bodies.

[7] A. Elzenaar, G. J. Martin, and J. Schillewaert, On thin Heckoid and generalised triangle groups in $\mathsf{PSL}(2,\mathbb{C})$, NUS/IMS Scientific Reports: Computational Aspects of Thin Groups, 03 Jun 2024–14 Jun 2024, 2024. arXiv 2409.04438 [math.GT].

Abstract: We provide a brief overview of our upcoming work identifying all the thin Heckoid groups in $\mathsf{PSL}(2,\mathbb{C})$. Here we give a complete list of the 55 thin generalised triangle groups of slope $1/2$. This work was presented at the conference Computational Aspects of Thin Groups, IMSS, Singapore and presents an application of joint work initiated with Colin Maclachlan.

[8] A. Elzenaar, J. Gong, G. J. Martin, and J. Schillewaert, Bounding deformation spaces of Kleinian groups with two generators, 2024, Updated Jun 2026. arXiv 2405.15970 [math.CV].

Abstract: In this article we provide simple and provable bounds on the size and shape of the locus of discrete subgroups of $\mathsf{PSL}(2,\mathbb{C}) \simeq \mathrm{Isom}^+(\mathbb{H}^3)$ which split as a free product of cyclic groups $\mathbb{Z}_p * \mathbb{Z}_q$, $3 \leq p, q \leq \infty$. These bounds are sharp and meet the highly fractal boundary of the deformation space in four cusp groups. Such bounds have great utility in computer assisted searches for extremal Kleinian groups so as to identify universal constraints (volume, length spectra, etc.) on the geometry and topology of hyperbolic 3-orbifolds. As an application, we prove a strengthened version of a conjecture by Morier-Genoud, Ovsienko, and Veselov, motivated by the theory of quantum rational numbers, on the faithfulness of the specialised Burau representation.

Published

[9] A. Elzenaar, G. J. Martin, and J. Schillewaert, The combinatorics of Farey words and their traces, *Groups, Geometry, and Dynamics* **20** no. 1 (2026), 215–256. MR 5032293. Zbl 8166380. arXiv 2204.08076 [math.GT]. `https://doi.org/10.4171/GGD/832`.

Abstract: We introduce a family of 3-variable 'Farey polynomials' that are closely connected with the geometry and topology of 3-manifolds and orbifolds as they can be used to produce concrete realisations of the boundaries and local coordinates for one-dimensional (over $\mathbb{C}$) deformation spaces of Kleinian groups. As such, this family of polynomials has a number of quite remarkable properties. We study these polynomials from an abstract combinatorial viewpoint, including a recursive definition extending that which is known in the literature for the special case of manifolds, even beyond what the geometry predicts. We also present some intriguing examples and conjectures which we would like to bring to the attention of researchers interested in algebraic combinatorics and hypergeometric functions. The results in this paper additionally provide a practical approach to various classification problems for rank 2 subgroups of $\mathsf{PSL}(2,\mathbb{C})$ since they, together with other recent work of the authors, make it possible to provide certificates that certain groups are discrete and free, and effective ways to identify relators.

[10] A. Elzenaar and S. Waldron, Putatively optimal projective spherical designs with little apparent symmetry, *Journal of Combinatorial Designs* **33** no. 6 (2025), 222–234. MR 4895104. Zbl 1562.05025. arXiv 2405.19353 [math.CO]. `https://doi.org/10.1002/jcd.21979`.

Abstract: We give some new explicit examples of putatively optimal projective spherical designs. i.e., ones for which there is numerical evidence that they are of minimal size. These form continuous families, and so have little apparent symmetry in general, which requires the introduction of new techniques for their construction. New examples of interest include an 11-point spherical $(3,3)$-design for $\mathbb{R}^3$, and a 12-point spherical $(2,2)$-design for $\mathbb{R}^4$ given by four Mercedes-Benz frames that lie on equi-isoclinic planes. We also give results of an extensive numerical study to determine

the nature of the real algebraic variety of optimal projective real spherical designs, and in particular when it is a single point (a unique design) or corresponds to an infinite family of designs.

[11] A. Elzenaar, G. J. Martin, and J. Schillewaert, Concrete one complex dimensional moduli spaces of hyperbolic manifolds and orbifolds, in *2021–22 MATRIX Annals* (D. R. Wood, J. de Gier, and C. E. Praeger, eds.), Springer, 2024, pp. 31–74. MR 4807267. Zbl 1520.30060. arXiv 2204.11422 [math.GT].

Abstract: The Riley slice is arguably the simplest example of a moduli space of Kleinian groups; it is naturally embedded in $\mathbb{C}$, and has a natural coordinate system (introduced by Linda Keen and Caroline Series in the early 1990s) which reflects the geometry of the underlying 3-manifold deformations. The Riley slice arises in the study of arithmetic Kleinian groups, the theory of two-bridge knots, the theory of Schottky groups, and the theory of hyperbolic 3-manifolds; because of its simplicity it provides an easy source of examples and deep questions related to these subjects. We give an introduction for the non-expert to the Riley slice and much of the related background material, assuming only graduate level complex analysis and topology; we review the history of and literature surrounding the Riley slice; and we announce some results of our own, extending the work of Keen and Series to the one complex dimensional moduli spaces of Kleinian groups isomorphic to $\mathbb{Z}/a\mathbb{Z} * \mathbb{Z}/b\mathbb{Z}$ acting on the Riemann sphere, $2 \leq a, b \leq \infty$. The Riley slice is the case of two parabolic generators where $a = b = \infty$.

[12] A. Elzenaar, G. J. Martin, and J. Schillewaert, Approximations of the Riley slice, *Expositiones Mathematicae* **41** no. 1 (2023), 20–54. MR 4557273. Zbl 1520.30060. arXiv 2111.03230 [math.GT]. https://doi.org/10.1016/j.exmath.2022.12.002.

Abstract: Adapting the ideas of L. Keen and C. Series used in their study of the Riley slice of Schottky groups generated by two parabolics, we explicitly identify 'half-space' neighbourhoods of pleating rays which lie completely in the Riley slice. This gives a provable method to determine if a point is in the Riley slice or not. We also discuss the family of Farey polynomials which determine the rational pleating rays and their root set which determines the Riley slice; this leads to a dynamical systems interpretation of the slice. Adapting these methods to the case of Schottky groups generated by two elliptic elements in subsequent work facilitates the programme to identify all the finitely many arithmetic generalised triangle groups and their kin.

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